

Contestant Number: _____

Time: _____

Rank: _____



LINUX OPERATING SYSTEM FUNDAMENTALS ~PILOT~ (350)

STATE – 2020

TOTAL POINTS

_____ (500 Points)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

TEST TIME: 60 Minutes

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1. Which of the following commands should a Linux user enter to see a list of all the commands the user recently ran at the command prompt?
 - A. history
 - B. clear
 - C. chsh
 - D. uname

2. Which of the following is displayed when the **uname -a** command is run?
 - A. All system information
 - B. The current working directory
 - C. The current username
 - D. The names of files and directories in the current directory

3. Which of the following commands searches man pages for a specific keyword?
 - A. man -s
 - B. find
 - C. slocate
 - D. apropos

4. You want the directory /sbin/special to be include in the PATH environment variable. You also want to keep all the current directory entries currently in the PATH variable. Which of the following commands would you use?
 - A. PATH=\$PATH:\$/sbin/special
 - B. PATH=/sbin/special
 - C. PATH=\$PATH:/sbin/special
 - D. PATH=PATH&/sbin/special

5. Anna, a system administrator, wants to set a global variable that can be used each time she logs in. Currently, she has to set the variable each time a terminal is opened. Which of the following files would need to be modified to make a global variable persistent?
 - A. ~/.bash_profile
 - B. ~/.bash_history
 - C. /etc/admin_profile
 - D. ~/.bash_logout

6. Which of the following commands sorts the combined contents of the wordlist1 and wordlist2 files and sends the results to both the screen and a file named sortedwordlist?
 - A. cat /usr/wordlist1 /usr/wordlist2 | tee sortedwordlist
 - B. cat /usr/wordlist1 /usr/wordlist2 | sort | tee sortedwordlist
 - C. cat /usr/wordlist1 /usr/wordlist2 | sort sortedwordlist
 - D. cat /usr/wordlist1 >> /usr/wordlist2 | sort sortedwordlist

7. Which of the following commands utilize command substitution?

- A. `echo -e "sort -r names.txt"`
- B. `echo -s "sort -r names.txt"`
- C. `myvar=$(sort -r names.txt)`
- D. `myvar="echo 'sort -r names.txt'"`

8. Within the `/var` directory is a subdirectory named `backup`. Ken, a Linux administrator, needs to delete the directory `backup` and any files that it contains. He changes directory focus to `/var`. Which of the following commands deletes the directory named `backup` and any file it contains?

- A. `rm backup/*`
- B. `rm backup`
- C. `rm -r backup/*`
- D. `rm -r backup`

9. A Linux administrator is logged in as `root` and needs to copy a file named *letter.doc* from the USB flash drive mounted under `/media/usb`. Which of the following commands will copy the file from the USB flash drive to the `root` user's home directory?

- A. `cp /media/usb/letter.doc home:`
- B. `cp /media/usb/letter.doc /root`
- C. `cp /dev/usb/letter.doc /root`
- D. `cp /mnt/usb:letter.doc /root`

10. According to the Filesystem Hierarchy Standard (FHS), which of the following directories contains information about the system state and processes?

- A. `/proc`
- B. `/tmp`
- C. `/bin`
- D. `/root`

11. Which of the following commands finds files with the `.txt` extension in the `/home/gshant` directory?

- A. `find /home/gshant -name '*.txt'`
- B. `find /home/gshant -name '*txt*'`
- C. `find /home/gshant -type f -name '*text*'`
- D. `find /home/gshant -type d -name '*text*'`

12. Which of the following commands is used to find a specific file on a Linux system?

- A. `whereis`
- B. `search`
- C. `type`
- D. `locate`

13. Which of the following commands displays all the lines in the blue_and_gold file that do not contain the word 'Karen'?

- A. `grep -n blue_and_gold Karen`
- B. `grep -n Karen blue_and_gold`
- C. `grep -v Karen blue_and_gold`
- D. `grep -v blue_and_gold Karen`

14. An application named ABCD is generating system errors when it starts. Which of the following commands searches the system message log file for these errors?

- A. `tail -n25 /var/log/messages`
- B. `tail -n25 /var/log/ABCD.log`
- C. `top /var/log/ABCD.log`
- D. `grep ABCD /var/log/messages`

15. Which of the following directories is not required to be part of the partition that holds the root (/) directory?

- A. `/bin`
- B. `/etc`
- C. `/usr`
- D. `/proc`

16. Which of the following character sets is a variable length encoding standard of Unicode that uses one to four 8-bit bytes to support many languages and most commonly used on Linux?

- A. ASCII
- B. UTF-8
- C. LANG
- D. LC_ALL

17. Which of the following commands will help you to determine which version of GRUB is installed on your Linux system?

- A. `grub2-install --version`
- B. `grub-install -migrate`
- C. `grub1.98-install -V`
- D. `update-grub`

18. You are working on a systemd-based Linux distribution. Which command can you use to manage the services and targets?

- A. `systemctl`
- B. `initctl`
- C. `telinit`
- D. `insserv`

19. Which of the following are examples of a Linux display server?

- A. KDE
- B. X11
- C. Unity
- D. KWin

20. Which of the following desktops was derived from GNOME 3, follows traditional desktop metaphor conventions, and is the primary desktop environment for the Mint distribution?

- A. Cinnamon
- B. Unity
- C. KDE
- D. MATE

21. Linux provides technology that allows the user to remotely view a graphical user interface and control mouse and keyboard events. Which of the following technologies establishes a connection with a remote computer using a viewer on one end and a server on the remote end?

- A. XRDP
- B. SPICE
- C. SRDP
- D. VNC

22. You need to install the package `apt-0.5.15cnc6-1.1.fc2.fr.i386.rpm`. Which of the following commands will perform the installation? (Select TWO).

- A. `rpm -Uh apt-0.5.15cnc6-1.1.fc2.fr.i386.rpm`
- B. `rpm -e apt-0.5.15cnc6-1.1.fc2.fr.i386.rpm`
- C. `rpm -Xh apt-0.5.15cnc6-1.1.fc2.fr.i386.rpm`
- D. `tar -Uh apt-0.5.15cnc6-1.1.fc2.fr.i386.rpm`

23. You have an RPM package called *mathpac* that is not working correctly. You have downloaded the latest package. Which of the following commands should you use to install the latest mathpac package?

- A. Run: `rpm -e mathpac` Followed by: `rpm -i mathpac`
- B. `rpm -U mathpac`
- C. `rpm -F mathpac`
- D. `rpm -i mathpac`

24. You have installed a package called *mathpac* with DNF. After a system upgrade, the package is not working correctly. Which of the following commands should you try first to get the package to function?

- A. `dnf remove mathpac` and then `dnf install mathpac`
- B. `rpm -U mathpac`
- C. `dnf update mathpac`
- D. `dnf upgrade mathpac`

25. Which **dpkg** option completely removes a package and its configuration files?

- A. `-R`
- B. `-r`
- C. `-P`
- D. `--configure`

26. Which if the following directories typically hold the dynamic libraries on your system?

- A. `/usr/lib/`
- B. `/usr/remote/lib/`
- C. `/etc/ld.so.conf/`
- D. `/lib/ld.so/`

27. A file contains the following entry:

`sales:x:1001:pclark,mmckay,hsamson`

Which of the following files contains similar entries?

- A. `/etc/group`
- B. `/etc/shadow`
- C. `/etc/gshadow`
- D. `/etc/passwd`

28. You are viewing the `/etc/passwd` file, and you notice the following entry:

`pclark:x:1001:1001:Petunia Clark:/home/pclark:/bin/bash`

What statement BEST describes this entry?

- A. The `pclark` password is the letter `x`.
- B. The `pclark` password is stored in the `/etc/shadow` file.
- C. The `pclark` user account is locked.
- D. The `pclark` user account has not set a password.

29. A user with an account name *larry* has just been terminated from the company. There is good reason to believe that the user will attempt to access and damage files in your system in the very near future. Which command below will disable or remove the user account from the system and remove his home directory?

- A. `userdel larry`
- B. `userdel -r larry`
- C. `userdel -home larry`
- D. `userdel -h larry`

30. Which of the following is the maximum number of primary partitions that can be created on a single hard disk drive?

- A. 2
- B. 4
- C. 8
- D. Unlimited

31. Which of the following commands creates a logical volume named *video* from the *home* volume group and configures it with 2 TB of disk space?

- A. `lvcreate -L 2T -n video home`
- B. `lvcreate -L 2T -n home video`
- C. `vgcreate -L 2T -n video home`
- D. `vgcreate -L 2T -n home video`

32. Which file shows the currently mounted volumes?

- A. `/etc/fstab`
- B. `/etc/mount`
- C. `/etc/mount.conf`
- D. `/etc/mtab`

33. When creating an ext2 or ext3 file system, a fixed number of inodes is assigned. Which of the following commands shows the number of inodes, number of free inodes, and number of used inodes?

- A. `du --inodes`
- B. `ls -i`
- C. `mke2fs -i`
- D. `df -i`

34. You must change ownership of a script called *myscript* so that only the user owner has read/write access to it and only the user owner and group owner have execute permissions. Which of the following commands will accomplish this task?

- A. `chmod 750 myscript`
- B. `chmod 771 myscript`
- C. `chmod 611 myscript`
- D. `chmod 755 myscript`

35. A file named *acctg* needs to have the SGID permission set on it while keeping all other permissions at their present value. Which of the following commands will accomplish this task?

- A. `chmod +g acctg`
- B. `chmod s+g acctg`
- C. `chmod g+s acctg`
- D. `chmod 1777 acctg`

36. You need to back up Jane's files in her `/home/user/jane` directory. Which command will create an archive of these files in the `/root` directory?

- A. `tar -cvf /root/jane.tar /home/user/jane`
- B. `tar -cjvf jane.tbz /home/user/jane`
- C. `tar -cbvf /root/jane.tbz /home/user/jane`
- D. `tar -cgvf /root/jane/tgz /home/user/jane`

37. You want to view all currently-running processes that include *getty* in the process name. Which command would you use?

- A. `ps -A | grep getty`
- B. `ps -a | grep getty`
- C. `ps -a | find getty`
- D. `ps -A | find getty`

38. Which file should you edit to schedule a task to execute every week on Saturday?

- A. `/etc/cron.monthly`
- B. `/etc/cron.weekly`
- C. `/etc/cron.daily`
- D. `/etc/cron.hourly`

39. A Linux system can determine the time and date in which of the following ways?

- A. It can use NNTP to set the time.
- B. It can use SMTP to set the time.
- C. It can use FTP to set the time.
- D. It can use NTP to set time.

40. You are an application developer. You are writing a program to exchange video files through a TCP/IP network. You need to select a transport protocol that will guarantee delivery. Which TCP/IP protocol would you implement that provides this capability?

- A. TCP
- B. UDP
- C. IP
- D. RIP

41. A user calls the help desk to report that she is unable to connect to the network after an upgrade of her Linux operating system. You suspect the problem is related to the network configuration. Which of the following commands could you use to quickly see the current IP address(es) assigned to her computer?

- A. netstat -a
- B. host
- C. ifconfig -a
- D. network -a

42. Which of the following is a valid way to assign a variable a value in a bash script?

- A. num1==5
- B. variable1=Hello
- C. num1 := 7;
- D. type string variable1=Hello

43. Given the following bash script:

```
#!/bin/bash
declare -i count=5
until [ $count -lt 3 ]
do
    echo count $count
    count=count-1
done
```

Which of the following shows the output from this script?

- A. count 5 count 4 count 3
- B. count 5 count 4
- C. count 1 count 2 count 3 count 4 count 5
- D. count 3 count 4 count 5

44. What does the **netstat -a** command show?

- A. All listening sockets
- B. All connected hosts
- C. All listening and non-listening sockets
- D. All network users

45. You would like to make it harder for malicious users to gain access to sensitive information. Which of the following techniques can be used to remap the root directory to include only certain directories and files?

- A. SSH certificate
- B. PKI
- C. chroot jail SSH
- D. One-time password

46. When working with Linux, it is important to understand what and how distributions are used. Which of the following BEST describes a Linux distribution?

- A. A pre-written code element programmers can use within programs.
- B. The kernel used to build a Linux operating system.
- C. A unique compilation of the Linux operation system.
- D. An application capable of running on Linux.

47. Your company is running a critical business application. The executive team wants to ensure the server is available at all times, even in the event of a server failure. Which of the following server roles would be used to provide a failover server in the event of a system failure?

- A. Clustering
- B. Load balancer
- C. Proxy
- D. SNMP

48. Which of the following is a multifactor authentication method supported by Linux?

- A. Mantrap
- B. Kerberos
- C. Fingerprint
- D. TACACS+

49. Which of the following shell declarations should be entered on the first line of a script for a system that uses the bash shell?

- A. `/bin/bash`
- B. `#!/bin/bash`
- C. `/bin/tsh`
- D. `#!/bin/csh`

50. You have created a separate partition for the sales team and mounted it to the `/sales` directory. The team is about to begin a new project, and they want to make sure there is enough disk space to hold upcoming files. Which of the following commands will show you the amount of free space?

- A. `diff /sales`
- B. `free /sales`
- C. `part /sales`
- D. `df /sales`